

Nikhil Maroju

Data Scientist

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Summary

Results-driven Data Scientist with 4+ years of experience in building and deploying machine learning and deep learning models. Expertise in data processing, feature engineering, model tuning, and cloud-based deployment using AWS and Azure platforms. Skilled in Python, PySpark, and various ML frameworks, with a strong focus on improving model accuracy, scalability, and operational efficiency. Adept at collaborating with cross-functional teams to deliver data-driven solutions that drive business value.

Technical Skills

- **Programming & Frameworks:** Python, R, PySpark, Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Flask, Docker, Kubernetes
- **Cloud Platforms & Services:** AWS (SageMaker, Lambda, Glue, S3), Azure (ML Pipelines, Databricks, Data Lake, Kubernetes), GCP
- **Data Engineering & Big Data:** SQL, Pandas, Azure Data Factory, Apache Spark, Hadoop, Apache Airflow, ETL Processes
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Neural Networks, Autoencoders, Anomaly Detection, Time-Series Forecasting, Reinforcement Learning
- **Natural Language Processing (NLP):** Text Preprocessing, Sentiment Analysis, Named Entity Recognition (NER), Transformers
- **Data Visualization & BI Tools:** Matplotlib, Seaborn, Tableau, Power BI, Plotly, Neo4j (Graph Analytics)
- **Model Deployment & MLOps:** MLflow, CI/CD Pipelines, Model Versioning, Hyperparameter Tuning (Optuna), A/B Testing, Model Explainability (SHAP, LIME)
- **Statistical Analysis & Mathematical Modeling:** Statistical Inference, Hypothesis Testing, Linear Algebra, Multivariate Analysis
- **Soft Skills:** Problem-Solving, Critical Thinking, Communication, Collaboration, Adaptability, Continuous Learning, Ethical Judgment

Professional Experience

Data Scientist, Deutsche Bank

10/2024 – Present | Remote, USA

- Worked on Anti-Money Laundering (AML) Pattern Detection, collaborated with compliance, risk, and legal teams, led requirement-gathering sessions to define typologies, and used semi-supervised deep learning to detect hidden money laundering behaviors in complex transaction flows.
- Collected and processed raw financial data with data engineers via AWS Glue and PySpark, and collaborated with data analysts to clean, normalize, and shape millions transactions for training AI models using SQL and Pandas.
- Built a Deep Autoencoder model in PyTorch to capture abnormal patterns in transaction behavior, performed advanced feature engineering, and trained the model with 70-30 split to distinguish typical activity from rare high-risk outliers.
- Validated model using reconstruction error metrics and clustering visualization, applied A/B testing across different threshold levels, and improved detection precision by 24% while reducing false alert rates in operational monitoring systems by 40%.
- Performed hyperparameter tuning using Bayesian optimization with Optuna and AWS SageMaker, adjusting encoder depth and learning rates, which enhanced detection accuracy by 21% across domestic and cross-border payment scenarios.
- Deployed the model via AWS Lambda with support from DevOps, built REST endpoints, integrated Neo4j-based visual analytics for entity relationship tracking, and enabled real-time flagging that increased compliance efficiency by 33% across AML teams.

Data Scientist, Mu Sigma

01/2021 – 08/2023 | Hyderabad, India

- Spearheaded the development of a Customer 360 analytics solution by integrating customer data across CRM, e-commerce, and loyalty platforms, improving personalization capabilities and driving a 25% increase in targeted campaign ROI.
- Led data wrangling and pipeline automation using PySpark and Azure Data Factory, enabling ingestion and transformation of 5 TB/month of structured and semi-structured data into Azure Data Lake with >98% data quality.
- Engineered complex features such as customer lifetime value (CLV), churn propensity, and RFM scores, resulting in a 32% boost in segmentation model accuracy for marketing teams.
- Built and evaluated machine learning models using Scikit-learn and XGBoost for customer churn and cross-sell predictions, achieving AUC-ROC of 0.91 and F1-score of 0.78, with continuous model tracking using MLflow.
- Tuned hyperparameters via Optuna, implemented k-fold cross-validation, and used Azure ML pipelines to operationalize retraining and evaluation processes, increasing model lifecycle efficiency by 45%.
- Containerized model APIs using Flask and Docker, and deployed to Azure Kubernetes Service (AKS) through CI/CD pipelines, reducing scoring latency by 50% and enabling real-time decisioning for marketing and CX teams.
- Delivered comprehensive model explainability using SHAP and LIME, ensuring transparency and interpretability for both business users and regulatory compliance teams.

Jr. Data Scientist, Mindtree Ltd

01/2020 – 12/2020 | Hyderabad, India

- Assisted in the development of a predictive maintenance solution for a manufacturing client, using IoT sensor data and Azure Databricks to reduce machine downtime by 30% through anomaly detection and time-series forecasting.
- Engineered features from large-scale streaming data and built XGBoost models, improving failure prediction accuracy by 35%, with hyperparameter tuning via Optuna and continuous tracking using MLflow.
- Deployed model APIs on Azure Kubernetes Service with Docker and Flask, enabling real-time alerts and dashboard visualizations, boosting operational efficiency and reducing unplanned maintenance costs.

Education

Indiana Wesleyan University — Marion, IN, USA

Master of Information Technology

09/2023 – 05/2025